

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and debugged embedded C/C++ firmware for ESP32 microcontrollers, implementing **REST APIs** and **MQTT** for cloud IoT integration, maintaining **99.9%** message delivery success rate.
- Engineered robust communication stacks for **Bluetooth (BLE)** and **Wi-Fi** provisioning, optimizing memory footprints by **15%** on constrained devices and managing connection stability.
- Designed and implemented **Modbus RTU** communication drivers over **RS-485**, successfully integrating with industrial inverters and reducing data acquisition latency by **20%**.
- Authored Python scripts for automated testing and device provisioning, cross-compiling toolchains, and managing serial interfaces within Linux environments, accelerating deployment cycles by **30%**.
- Managed production incident response for enterprise applications, analyzing logs via **Splunk** and **Dynatrace**, and coordinating resolution efforts to meet stringent SLAs.
- Contributed to the development of enterprise business systems, performing change-control management for system configurations and coordinating User Acceptance Testing (UAT) across interconnected platforms.

SKILLS

Embedded Systems – Embedded C/C++, ESP32, ESP-IDF, RTOS, FreeRTOS, Microcontrollers, Bare-metal development

Wireless & Wired Protocols – Bluetooth (BLE), Wi-Fi, TCP/IP, Modbus (RTU/TCP), RS-485, I2C, UART, Matter, Thread

Cloud IoT & Security – MQTT, REST APIs, WebSockets, Protobuf, AWS IoT Core, Device Provisioning, TLS, Public-key cryptography, Secure boot

Software Development & Tools – Python, Linux, Shell scripting, Git, Jenkins, CI/CD, Docker, Cross-compilation

Testing & Documentation – Unit testing, Integration testing, Firmware testing, Technical documentation

Enterprise Systems – Java Spring Boot, PostgreSQL, MySQL, Oracle, ERP integration, Change Management

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Smart Home Device Connectivity Framework

- Developed firmware for an ESP32-based smart home hub using **Embedded C++**, integrating **BLE** and **Wi-Fi** connectivity for device discovery and control.

- Implemented **MQTT** client for seamless communication with a cloud IoT platform, enabling real-time data streaming and remote command execution with **99.8%** reliability.
- Authored Python scripts to automate firmware flashing and provisioning for a fleet of **100+** devices, reducing setup time by **40%**.

Industrial Automation Data Gateway

- Created a gateway application in C++ running on a Linux-based embedded system, interfacing with industrial sensors via **Modbus TCP** and **I2C**.
- Designed a data abstraction layer to translate proprietary sensor readings into standardized JSON payloads for a backend analytics service.
- Integrated secure communication channels using **TLS**, ensuring data integrity and authenticity for sensitive industrial operational data.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)